SAMARTHA RAYGURU

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Education

Vivekanand Education Society's Institute Of Technology Jun. 2024
B.E. in Information Technology, Minor in Data Science ***CGPA: 9.34***

CFA Level I Candidate, Aug. 2026 Exam

Relevant Coursework: Data Structures & Algorithms, Big Data Analytics, Artificial Intelligence, Information Retrieval Systems, Probability & Statistics
Technical Skills: Python, SQL, Excel, Power BI, pandas, Streamlit, Machine Learning, Data Modeling

Experience

Barclays Bank PLC

Technology Developer, BA4 *Mar. 2025 - Present*

- Lead validation of trade and book-level risk datasets for completeness, reconciliation, and integrity.
- Investigate data breaks with stakeholders, perform root-cause analysis, and drive process optimisations.
- Used LLM tooling to accelerate root-cause analysis on data breaks and surface anomaly patterns earlier.

Technology Graduate Developer, BA3 *Jul. 2024 - Mar. 2025*

- Supported risk data lifecycle for trade and book-level datasets, focusing on exposure metrics (EAD) and sensitivities.
- Reconciled datasets against upstream feeds; validated notional, counterparty, exposure, and sensitivity fields.
- Investigated recurring data breaks, traced causes to upstream issues, and refined controls to reduce exceptions.
- Built monitoring views flagging anomalies and threshold breaches across counterparty cuts.

Barclays Bank PLC Jun. 2023 - Aug. 2023

BA3 Intern, Markets Post Trade (US Cash Settlements)

- Built a settlement monitoring dashboard (React, Oracle DB, Kibana on ELK stack) tracking fails, ageing, and counterparty exposure.
- Designed KPI cards and threshold-based alerts to flag operational and counterparty exceptions earlier.
- Built concentration views by counterparty and geography with top-N cuts and a green, amber, red watchlist on ageing.

Projects

Borrower Default Prediction Model Jun. 2025

Credit Risk Modeling Project *Python, scikit-learn, SHAP*

- Built ML models (logistic regression, random forest, XGBoost) on the German Credit dataset for default prediction.
- Evaluated models with ROC-AUC, KS statistic, and confusion matrices; tuned thresholds for approve/decline rules.
- Used SHAP to interpret predictions and prototyped an LLM rewriting outputs into plain-language reason codes.

Security Analysis & Portfolio Optimization Dec. 2024

Quantitative Portfolio Construction Project *Excel, Bloomberg, Mean-Variance Optimization*

- Optimised a multi-asset portfolio using Markowitz Mean-Variance with efficient-frontier risk-return analysis.
- Performed sensitivity analysis on asset weights and Sharpe ratios to assess stability under changing market conditions.

NIFTY 50 Sector Performance Analysis Mar. 2025

Equity Market Analysis Project *SQL (PostgreSQL), Python (pandas), Power BI*

- Pulled five years of NIFTY 50 prices into PostgreSQL; used SQL window functions and CTEs for returns and volatility.
- Computed annualised return, volatility, Sharpe, and rolling beta per sector with a cross-sector correlation matrix.
- Visualised findings in a Power BI dashboard with sector heatmaps, drawdown charts, and return distributions.

Achievements

Hack-A-Difference Hackathon, Barclays (Global Level) Jan. 2026

- Secured Top 2 globally for building a data-driven solution for **Magic Bus**, focused on intelligent document processing and analytics.

Core EOD Risk Hackathon, Barclays Dec. 2025

- Achieved 3rd place for designing a solution to enhance end-of-day risk data validation and reconciliation workflows.